



UNIVERSITY OF
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Superconductivity

Lecture 08 – “Gradient”

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Previously, GL free energy

We constructed a form of the free energy density

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 \quad (1)$$

We found the free energy minimum condition

$$f_s - f_n|_{|\psi|^2 = -a/b} = -\frac{1}{2b} \left(\frac{da}{dT} \right)^2 (T_c - T)^2, \quad (2)$$

The result may be summarised as

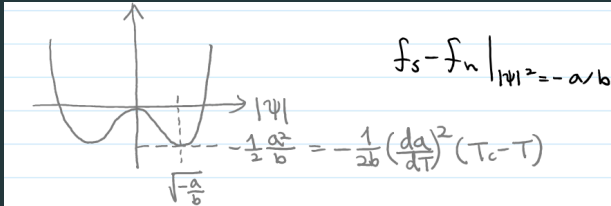


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Spatial Inhomogeneity

Adding a gradient term

What does it mean:

$$\psi \Rightarrow \psi(\mathbf{r}) \quad (3)$$

How to add $\nabla\psi(\mathbf{r})$ contribution to

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4. \quad (4)$$

One may consider $|\nabla\psi(\mathbf{r})|$, $|\nabla\psi(\mathbf{r})|^2$, $|\nabla\psi(\mathbf{r})|^3$, ..., but may keep one of the lowest order term, $|\nabla\psi(\mathbf{r})|^2$, and writes

$$f_s - f_n = a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{\hbar^2}{2m^*}|\nabla\psi(\mathbf{r})|^2, \quad (5)$$

One can guess why such coefficients are chosen (i.e., kinetic energy). Note that m^* is introduced as a parameter, which is not necessarily the mass of an electron.

Free energy minimum

Needs to find a free energy minimum, but note the change

$$f(T) \Rightarrow f(T, \mathbf{r}). \quad (6)$$

Needs to find the minimum for the total F than the density f as

$$F_s(T) = F_n(T) + \int \left(a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{\hbar^2}{2m^*}|\nabla\psi(\mathbf{r})|^2 \right) d^3r. \quad (7)$$

Condition: Ginzburg-Landau equation

How to find minimum?

- Looking for $\delta F_s(T) = 0$
- upon infinitesimal variation of $\psi(\mathbf{r})$, i.e., $\psi(\mathbf{r}) \rightarrow \psi(\mathbf{r}) + \delta\psi(\mathbf{r})$
- keep the lowest order of $\delta\psi(\mathbf{r})$ only

One arrives at the so-called Ginzburg-Landau equation,

$$-\frac{\hbar^2}{2m^*}\nabla^2\psi + a\psi + b\psi|\psi|^2 = 0, \quad (8)$$

Impression?

Looks like a Schrodinger equation (but with a non-linear term).

Normal-superconductor interface

Normal-superconductor interface (1/3)

Suppose a metal for $x < 0$ and a superconductor for $x > 0$:

$$-\frac{\hbar^2}{2m^*} \frac{d^2\psi(x)}{dx^2} + a\psi(x) + b\psi(x)|\psi(x)|^2 = 0, \quad (9)$$

for $x > 0$, where $\psi(0) = 0$.

Let's make this look simpler. Start by rearranging (and recalling $a < 0$),

$$\frac{\hbar^2}{2m^*|a|} \frac{d^2\psi(x)}{dx^2} + \psi(x) - \left(\frac{b}{|a|}\right) \psi(x)|\psi(x)|^2 = 0, \quad (10)$$

Also recall from the previous lecture that we obtained $|\psi|^2 = -a/b$ as a solution for a homogeneous case, which allows us to write

$$\psi(x \rightarrow \infty) \equiv \psi_\infty = \sqrt{\frac{|a|}{b}} \quad (11)$$

Normal-superconductor interface (2/3)

Then, through further defining

$$\xi(T) \equiv \left(\frac{\hbar^2}{2m^*|a(T)|} \right)^{1/2}, \quad (12)$$

$$y \equiv x/\xi, \quad (13)$$

and

$$\psi(x) = \psi_\infty f(y), \quad (14)$$

one can rewrite

$$\frac{\hbar^2}{2m^*|a|} \frac{d^2\psi(x)}{dx^2} + \psi(x) - \left(\frac{b}{|a|} \right) \psi(x)|\psi(x)|^2 = 0, \quad (15)$$

into a much simpler form,

$$-\frac{d^2}{dy^2}f(y) - f(y) + f^3(y) = 0. \quad (16)$$

Normal-superconductor interface (3/3)

The solution to

$$-\frac{d^2}{dy^2}f(y) - f(y) + f^3(y) = 0, \quad (17)$$

one can easily confirm is

$$f(y) = \tanh(y/\sqrt{2}). \quad (18)$$

Now, restore the original variables give

$$\psi(x) = \psi_\infty \tanh\left(\frac{x}{\sqrt{2}\xi(T)}\right). \quad (19)$$

Interpretation: The order parameter ψ is zero at the interface, but increases fast as entering into the superconductor and reaches the value expected for a uniform superconductor.

Coherence length

The Ginzburg-Landau coherence length,

$$\xi(T) \equiv \left(\frac{\hbar^2}{2m^* |a(T)|} \right)^{1/2}, \quad (20)$$

defines a scale over which ψ or n_s varies. One can replace a with $a(T) = \frac{da}{dT}(T - T_c)$, as obtained in the previous lecture, to obtain

$$\xi(T) = \left(\frac{\hbar^2}{2m^* \frac{da}{dT} (T - T_c)} \right)^{1/2} \quad (21)$$

$$= \left(\frac{\hbar^2}{-2m^* \frac{da}{dT} T_c} \frac{T_c}{T_c - T} \right)^{1/2} \quad (22)$$

$$= \xi(0) \left| \frac{T - T_c}{T_c} \right|^{-1/2} \quad (23)$$

$$= \xi(0) |t|^{-1/2}, \quad (24)$$

where t is called the reduced temperature.

Exercise

1. The order parameter, $\psi(x)$, of a superconducting wire satisfies the Ginzburg-Landau equation

$$-\frac{\hbar^2}{2m^*} \frac{d^2\psi}{dx^2} + a\psi + b|\psi|^2\psi = 0. \quad (25)$$

By considering a current-carrying solution, $\psi(x) = \psi_0 e^{ikx}$, derive a formula for the critical current density, J_c , which is the maximum value of the supercurrent density, $J_s = n_s^* e^* v_s$.